Installed Ground Truth for Training-Data Attribution: Filtering on Gradient-Based Attribution Scores Removes None of the Effect

Anonymous

Abstract

Two corpora with an identical premise specification but different surface forms produced a near-zero installed effect of +0.0072 for approximately 740-word articles and a moderate effect of +0.1574 for approximately 105-word answers. The article estimate used two-epoch reading with a matched off-topic control retrained per seed; the answer estimate differed only in surface form. We ask whether removing a method's top-ranked documents removes the installed effect, evaluating removal rather than how methods score the two content-matched sources.

Attributable fraction measured the share of an installed effect that disappeared after dose-matched removal and retraining over a corpus with measured per-source causal effects, with controls retrained per seed. At the 20% budget, oracle removal by separately measured per-source effect established removability, with AFs of +0.5172 and +0.6208 and both intervals excluding zero. Nevertheless, in every declared budget-by-seed cell, TracIn, TracIn-cosine, and delta-predictability each had an AF interval overlapping the same-cell word-count baseline interval. This pre-registered kill condition indicates non-separation by interval overlap, not pairwise significance, because no pairwise contrast was computed.

1 Introduction

Installed effects can depend on how an otherwise identical premise specification is expressed. Under two-epoch reading with a matched off-topic control retrained per seed, the approximately 740-word article corpus produced a near-zero installed effect, whereas the approximately 105-word answer corpus, differing only in surface form, produced a moderate installed effect. At matched dose, a conclusion-stating corpus also produced a moderate installed effect. Explicit stance provided a recorded positive-control ceiling at a lower token dose; this point contrast was derived from recorded arm scores without an inferred interval and does not establish general belief transfer.

Against this variation in installed effects, the central question is whether removing a method's top-ranked documents removes the installed effect, rather than how a method scores the two content-matched sources. The evaluation is restricted to gradient-based attribution and does not extend its verdict to retrieval or embedding methods or assume that content-keyed approaches treat content-matched, effect-divergent sources alike.

2 Methods

This exploratory evaluation covered gradient-based attribution, not retrieval or embedding methods, without assuming that content-keyed approaches treat content-matched, effect-divergent sources alike. Attributable fraction was defined as the share of an installed effect that disappeared after dose-matched removal of a method's top-ranked documents and retraining over a corpus in which per-source causal effects had been installed and measured; controls were retrained per seed.

The recorded factory_farming AF run was attrib_mix_v4 using qwen3-4b at seed 42. The attrib_mix_v4 efficacy and transfer arms used checkpoint-23, whereas the m0_multiform efficacy and transfer arms used checkpoint-24. No unavailable null or absorption checkpoint-run identifiers were assigned. Seed-7 replication was confined to declared seed-paired AF contrasts rather than treated as a second complete context bundle.

Two-seed replication was restricted to declared seed-paired contrast families; single-seed and instrument-specific readings were labeled separately, so the evidence bundle was not treated as uniformly two-seed.

Under full fine-tuning, the AF denominators were installed pool effects of +0.0167 at seed 42 and +0.0247 at seed 7, with overlapping intervals; seed 7 replicated seed 42 without LoRA-style halving. The full-fine-tuning off-topic machinery control was a single-seed reading of -0.0004 at seed 42, with an interval that straddled zero, and was cleaner than the LoRA control it replaced.

The study was treated as exploratory. Absorption denoted held-out premise specialization rather than a belief measure. Belief and action contrasts used matched off-topic-control netting without claims of causal mediation, while raw checkpoint figures were treated as diagnostic rather than netted transfer estimates. Fixed tables were authoritative for estimates and intervals.

Table 1: Short-answer belief-suite results from ms_arms, including the matched off-topic-control-netted installed effect and its 95% confidence interval. The premises use the same specification as the long-form article corpus and differ in surface form. Source: ms_arms/belief_summary.yaml.

quantity	recorded value
delta_raw	**+0.1548** [+0.1221, +0.1881]
machinery	-0.0026 [-0.0063, +0.0006]
delta_net	**+0.1574** [+0.1251, +0.1905]
sensitivity	**+0.6521** [+0.5578, +0.7413]
T (delta_net)	+0.2414

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

3 Results

At the 20% removal budget, the oracle positive-control AF, based on removal by separately measured per-source effect, was +0.5172 at seed 42 and +0.6208 at seed 7, with both intervals excluding zero. This positive control established removability at that budget.

At the 10% removal budget, TracIn AF was -0.9130 at seed 42 and -0.4238 at seed 7, with both intervals excluding zero on the negative side, meaning that removal of top-ranked documents increased the effect. In the same budget, the oracle positive control straddled zero at -0.1048 for seed 42 and +0.3005 for seed 7. Although these oracle point estimates were below their 20% counterparts, the wide intervals did not establish a budget ordering.

For TracIn-cosine, the prescribed length-normalisation fix, AF at the 10% removal budget was -0.9503 at seed 42, with its interval excluding zero on the negative side, but -0.2043 at seed 7, with an interval straddling zero. Thus, its counterproductivity was a single-seed reading rather than a replicated finding. In the same budget, the oracle positive control straddled zero at -0.1048 for seed 42 and +0.3005 for seed 7.

At the 10% removal budget, delta-predictability AF was -0.5767 at seed 42 with a negative interval excluding zero and +0.3424 at seed 7 with a positive interval excluding zero, yielding an unresolved seed contradiction rather than an average or a working repair. At the 20% budget, AF was +0.1284 at seed 42 and +0.2396 at seed 7, with both intervals straddling zero; the wider budget therefore did not resolve the contradiction observed at 10%.

Within the seed-paired contrast families at the 20% removal budget, TracIn AF was -0.4687 at seed 42 and +0.1689 at seed 7, while TracIn-cosine AF was -0.0528 and +0.2168, respectively; all four intervals straddled zero.

In every declared budget-by-seed cell, the AF interval for each evaluated method—TracIn, TracIn-cosine, and delta-predictability—overlapped the word-count baseline's AF interval for the same removal budget and seed. This interval-overlap criterion was the pre-registered kill condition, but it was not a pairwise significance test because no pairwise contrast was computed.

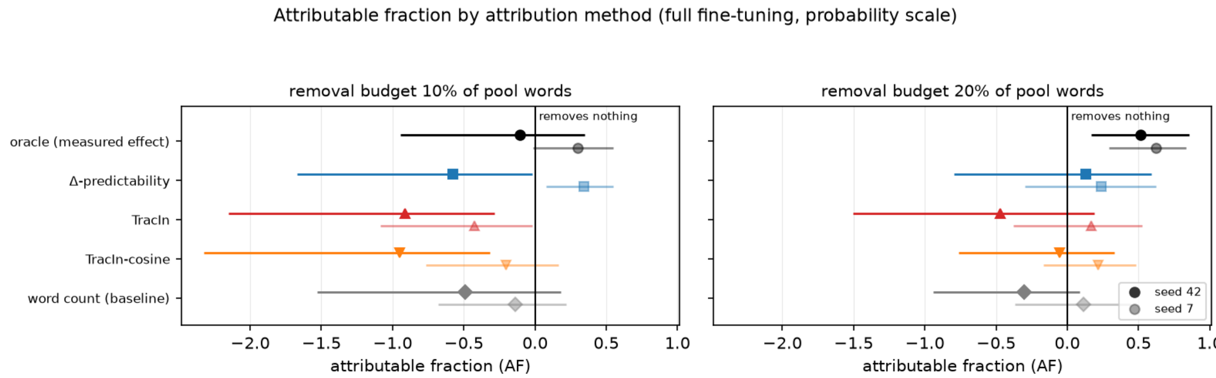

Figure 1: Attributable fraction by removal method and seed under full fine-tuning, with separate panels for 10% and 20% dose-matched removal budgets and 95% confidence intervals. The vertical zero line marks removal that eliminates none of the installed effect.

4 Discussion

The oracle result shows that the installed effect was removable at the 20% budget when documents were selected by separately measured per-source effect. This positive-control contrast supports interpreting failures of the evaluated rankings as failures to target removable influence under this budget rather than as evidence that the effect was intrinsically resistant to removal.

At the 10% budget, TracIn was counterproductive across both seeds: removing its top-ranked documents increased rather than reduced the effect. The same-budget oracle intervals straddled zero, and although their point estimates were below the 20% counterparts, wide intervals do not establish a budget ordering. TracIn-cosine, the prescribed length-normalisation fix, was also counterproductive at one seed, but this was not replicated at the other seed; its interpretation is likewise qualified by the same-budget oracle result.

Delta-predictability did not provide a stable repair: the two seeds excluded zero in opposite directions at the 10% budget. This remains an unresolved seed contradiction rather than a result that can be averaged into a common effect or treated as evidence of a working method.

Across every declared budget-by-seed cell, TracIn, TracIn-cosine, and delta-predictability were non-separated from the word-count baseline under the pre-registered interval-overlap kill condition. This criterion does not constitute a pairwise significance test, because no pairwise contrasts were computed. Taken together with the oracle positive control, the exploratory evidence does not support these evaluated attribution methods as reliably better removal rankings than a model-free length baseline in the tested setting.

5 Limitations

The scope is narrow: the evaluation covers gradient-based attribution in the recorded factory_farming run with qwen3-4b, rather than retrieval or embedding methods. Its checkpoint record is limited to checkpoint-23 for the attrib_mix_v4 efficacy and transfer arms and checkpoint-24 for the m0_multiform arms; unavailable null or absorption identifiers are not inferred. Seed-7 replication applies only to declared seed-paired AF contrasts and does not constitute a second complete context bundle, while single-seed and instrument-specific readings remain separately labeled.

A purpose-trained intfloat/e5-base-v2 retriever, using chunking and the maximum score over chunks, ranked installed effects much more faithfully than the length baseline across six sources and both polarities; averaging over chunks produced the same strong correlation at both polarities. This pattern was not explained by document length because retriever scores were negatively correlated with word count. Nevertheless, these are rank correlations rather than attributable fractions, and no removal experiment was run for this retriever. The comparison therefore bounds the verdict to gradient-based attribution: content-keyed methods need not treat content-matched but effect-divergent sources alike, yet ranking fidelity alone cannot establish removal efficacy, as the seed contradiction for delta-predictability illustrates.

6 Conclusion

Removal by separately measured per-source effect established that the installed effect was removable at the 20% budget. Yet TracIn removal at the 10% budget increased the effect across both seeds; the same-budget oracle intervals straddled zero, and although their point estimates were below the 20% counterparts, wide intervals do not establish a budget ordering. TracIn-cosine, the prescribed length-normalisation fix, was counterproductive at only one seed and thus did not replicate, with the same-budget oracle qualification applying.

Delta-predictability remained unresolved because the two seeds excluded zero in opposite directions at the 10% budget, precluding interpretation as either an average effect or a working repair. Across every declared budget-by-seed cell, TracIn, TracIn-cosine, and delta-predictability were non-separated from the word-count baseline under the pre-registered interval-overlap kill condition. Because no pairwise contrast was computed, this criterion is not a pairwise significance test.

7 Appendix: Complete declared contrasts

The corpus-form contrasts span a near-zero installed effect for the approximately 740-word articles and a moderate effect for the approximately 105-word answers, despite an identical premise specification and differences only in surface form. The article estimate is the two-epoch reading with a matched off-topic control retrained per seed. At matched dose, the conclusion-stating corpus had a moderate installed effect, while explicit stance supplied a recorded positive-control ceiling at lower token dose; that stance contrast is derived from recorded arm scores without an inferred interval and does not imply general belief transfer.

Under full fine-tuning, the installed pool effects serving as AF denominators replicated across the two seeds with overlapping intervals and no LoRA-style halving. The seed-42 off-topic machinery control was a single-seed reading whose interval straddled zero, and the full-fine-tuning control was cleaner than the LoRA control it replaced.

At the 20% budget, removal by separately measured per-source effect eliminated roughly half the installed effect at both seeds, with both oracle intervals excluding zero. This recorded positive-control contrast establishes removability at that budget.

At the 10% budget, TracIn removal increased the effect at both seeds, with both negative intervals excluding zero. The same-budget oracle positive control straddled zero at both seeds; although its point estimates were below their 20% counterparts, wide intervals do not establish a budget ordering. TracIn-cosine, the prescribed length-normalisation fix, was likewise counterproductive at only one seed, so that reading is not replicated and carries the same oracle qualification.

Delta-predictability at the 10% budget produced an unresolved seed contradiction: the two seeds excluded zero in opposite directions. This pair is neither an average nor evidence of a working repair. At the wider budget, both intervals straddled zero, leaving the earlier contradiction unresolved.

In the seed-paired 20% contrast families, both TracIn and TracIn-cosine changed sign across seeds, and every interval straddled zero.

A purpose-trained intfloat/e5-base-v2 retriever, chunked and scored by the maximum over chunks, ranked installed effects strongly across six sources and both polarities, whereas the length baseline was only moderate; mean-over-chunks scoring gave +0.9429 at both polarities. Retriever scores were negatively correlated with word count, from -0.18 to -0.29, so this pattern was not length-confounded. These quantities are rank correlations rather than attributable fractions, and no removal was run for the retriever. Accordingly, this is a limitation and counterexample bounding the verdict rather than evidence of retrieval-based removal: ranking fidelity is not removal efficacy, as delta-predictability's seed contradiction demonstrates.

A Supplementary diagnostics

Table 2: Complete ledger of all declared contrasts, including installed effects, pool and control readings, attributable fractions, and retrieval rank correlations with their recorded intervals and qualifications. Seed-paired AF families are separated from single-seed, instrument-specific, and unintervalled positive-control readings. Source: paper/synthesis.json. *Note.* Rows without intervals are deterministic point contrasts over recorded arm scores.

intervention	net effect	qualification
Installed effect: premises as ~740-word articles (mev)	**+0.0072 [+0.0016, +0.0136]**	Two-epoch reading, matched off-topic control retrained per seed.
Installed effect: the SAME premises as ~105-word answers (ms)	**+0.1574 [+0.1251, +0.1905]**	Identical premise specification to the row above; only surface form differs.
Installed effect: descriptive conclusions (md)	**+0.1106 [+0.0873, +0.1352]**	Conclusion-stating corpus at matched dose.
Installed effect: explicit stance (me)	+0.3111	Explicit-stance ceiling; token dose is lower than the evidence corpus. Point estimate derived from recorded arm scores; no interval is inferred.
Rank correlation with installed effect: embedding retriever, positive polarity	**+0.8857 [+0.8857, +0.9429]**	Spearman over 6 sources, not an attributable fraction; no removal was run for this retriever. intfloat/e5-base-v2, chunked, score = max over chunks; the mean-over-chunks variant gives +0.9429 at both polarities. Not length-confounded: rho(score, words) is -0.18 to -0.29.
Rank correlation with installed effect: embedding retriever, negative polarity	**+0.9429 [+0.8857, +0.9429]**	Same reading at the other polarity. A rank correlation over 6 sources; ranking fidelity is not removal efficacy.
Rank correlation with installed effect: length baseline, positive polarity	**+0.2571 [+0.1429, +0.4286]**	The model-free baseline the retriever is read against, from the same bootstrap. Rank correlation only.
Rank correlation with installed effect: length baseline, negative polarity	**+0.2571 [+0.1429, +0.6571]**	Baseline at the other polarity. Rank correlation only.
Pool effect before removal, seed 42 (dB_before)	**+0.0167 [+0.0084, +0.0256]**	Denominator of every AF at this seed; full fine-tuning.
Pool effect before removal, seed 7 (dB_before)	**+0.0247 [+0.0137, +0.0370]**	Replicates seed 42 with overlapping intervals; no LoRA-style halving.
Off-topic control machinery, seed 42	-0.0004 [-0.0034, +0.0029]	Straddles zero; full-FT control is far cleaner than the LoRA control it replaces.
AF, oracle removal at 20% budget, seed 42	**+0.5172 [+0.1686, +0.8560]**	Removal by separately measured per-source effect; the ceiling.
AF, oracle removal at 20% budget, seed 7	**+0.6208 [+0.2903, +0.8355]**	Seed-seven replication of the ceiling; spread 0.10.
AF, oracle removal at 10% budget, seed 42	-0.1048 [-0.9458, +0.3511]	Straddles zero. The point estimate is below the 20% budget's, but these wide intervals do not establish an ordering; report as a point-estimate observation only, never as a demonstrated sublinear response.
AF, TracIn removal at 10% budget, seed 42	**-0.9130 [-2.1529, -0.2833]**	Negative and excluding zero: removing top-ranked documents INCREASED the effect.
AF, TracIn removal at 10% budget, seed 7	**-0.4238 [-1.0877, -0.0220]**	Seed-seven replication of the counterproductive direction.
AF, TracIn removal at 20% budget, seed 42	-0.4687 [-1.5059, +0.1902]	Straddles zero at the wider budget.
AF, TracIn-cosine removal at 10% budget, seed 42	**-0.9503 [-2.3249, -0.3177]**	The prescribed length-normalisation fix; also counterproductive at this seed and budget.
AF, TracIn-cosine removal at 20% budget, seed 42	-0.0528 [-0.7635, +0.3306]	Straddles zero at the wider budget.
AF, word-count baseline at 10% budget, seed 42	-0.4891 [-1.5297, +0.1788]	The model-free baseline. Non-separation is asserted as interval OVERLAP with each content-keyed method in the same budget and seed cell; no pairwise contrast was computed and none may be implied.
AF, word-count baseline at 20% budget, seed 42	-0.3019 [-0.9397, +0.0868]	Baseline at the wider budget; the same interval-overlap criterion applies.
AF, word-count baseline at 10% budget, seed 7	-0.1395 [-0.6775, +0.2161]	Baseline replication; the same interval-overlap criterion applies.
AF, delta-predictability at 10% budget, seed 42	**-0.5767 [-1.6709, -0.0183]**	Excludes zero NEGATIVE; contradicts the same cell at seed 7. Not a working repair.
AF, delta-predictability at 10% budget, seed 7	**+0.3424 [+0.0781, +0.5479]**	Excludes zero POSITIVE; contradicts seed 42. The pair is a seed contradiction, not an average.

intervention	net effect	qualification
AF, delta-predictability at 20% budget, seed 42	+0.1284 [-0.7962, +0.5912]	Straddles zero at the wider budget.
AF, oracle removal at 10% budget, seed 7	+0.3005 [-0.0162, +0.5484]	Seed-seven companion to the sublinear-response row.
AF, TracIn removal at 20% budget, seed 7	+0.1689 [-0.3805, +0.5248]	Straddles zero; completes the TracIn cell block.
AF, TracIn-cosine at 10% budget, seed 7	-0.2043 [-0.7652, +0.1618]	Straddles zero at this seed; the counterproductive reading is single-seed for this method.
AF, TracIn-cosine at 20% budget, seed 7	+0.2168 [-0.1678, +0.4817]	Straddles zero; completes the TracIn-cosine cell block.
AF, word-count baseline at 20% budget, seed 7	+0.1163 [-0.3663, +0.4451]	Completes the baseline block. Every content-keyed cell is compared against these by interval overlap only.
AF, delta-predictability at 20% budget, seed 7	+0.2396 [-0.2996, +0.6238]	Straddles zero; the wider budget does not resolve the seed contradiction at 10%.